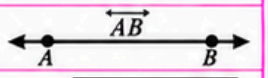
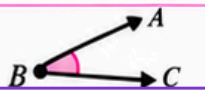
Geometry Basics Cheat Sheet

Unit Focus: points, lines, rays, segments, and angles

1 KEY VOCABULARY

point	An exact location in space.	
line	A straight path that extends forever in both directions.	
plane	A flat surface that extends forever in all directions.	
segment	A part of a line with two endpoints.	
ray	A part of a line with one endpoint that extends forever in one direction.	
angle	A figure formed by two rays with a common endpoint (vertex).	

collinear: points on the same straight line

coplanar: points on the same plane

2 NAMING AND DRAWING FIGURES

Use capital letters to name geometric figures.

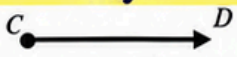
Line Segment



Segment $\overline{AB}$

$\overline{AB}$

Ray



Ray $\overrightarrow{CD}$

$\overrightarrow{CD}$

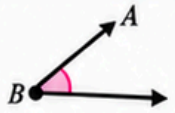
Line



Line EF

$\overleftrightarrow{EF}$

Angle



Angle ABC (vertex is B)

$\angle ABC$

IMPORTANT: Ray order matters.

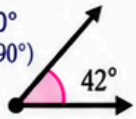
Ray CD differs from ray DC; angle ABC = CBA.

3 CLASSIFYING ANGLES

Angles are classified by their measures.

Acute Angle

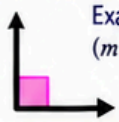
Less than 90° ($0° < m\angle < 90°$)



Example: 42° is an acute angle.

Right Angle

Exactly 90° ($m\angle = 90°$)



Obtuse Angle

Greater than 90° but less than 180° ($90° < m\angle < 180°$)



Straight Angle

Exactly 180° ($m\angle = 180°$)

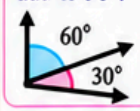


4 ANGLE RELATIONSHIPS

Special relationships between pairs of angles.

Complementary Angles

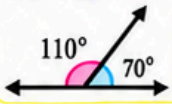
Two angles whose measures add to 90°.



$m\angle 1 + m\angle 2 = 90°$ (are complementary)

Supplementary Angles

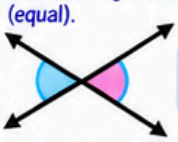
Two angles whose measures add to 180°.



$m\angle 1 + m\angle 2 = 180°$ (are supplementary)

Vertical Angles

Opposite angles formed by intersecting lines are congruent (*equal*).



Vertical angles are congruent.

Linear Pair

Two adjacent angles that form a straight line are supplementary (their measures add to 180°).

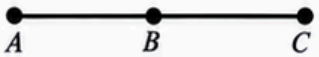


Example: If one angle in a linear pair is 68°, the other is 112°.

5 SEGMENT ADDITION AND MIDPOINT

Segment Addition Postulate

If point B is between A and C, then $AB + BC = AC$.



$AB + BC = AC$

Midpoint

A midpoint divides a segment into two congruent segments (equal lengths).



Example: If $AB = 18$ and M is the midpoint of AB, then $AM = MB = 9$.

6 USING ALGEBRA WITH ANGLES

Example: Solving for an angle measure

The angles $(x + 15)°$ and $(2x - 3)°$ are supplementary.

Step 1: Set up the equation.
$(x + 15) + (2x - 3) = 180$

Step 2: Combine like terms.
$3x + 12 = 180$

Step 3: Solve for x.
$3x = 168$
$x = 56$

Find the angle measures.

$x = 56$

First angle: $x + 15 = 56 + 15 = 71°$

Second angle: $2x - 3 = 2(56) - 3 = 112 - 3 = 109°$

Answer: The angles are 71° and 109°.

7 CONGRUENT FIGURES

Congruent segments have equal lengths.

Congruent angles have equal measures.

Use matching tick marks and angle arcs.

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Use degree symbols (°) with angle measures.
- Vertical angles are congruent (equal).
- A midpoint creates two equal segments.
- Decide whether angles should add to 90° or 180° (complementary or supplementary).
- Use correct notation and order when naming rays and angles.

COMMON MISTAKES:

- Forgetting to use degree symbols.
- Thinking vertical angles are supplementary (they are congruent).
- Not dividing a segment into equal parts at the midpoint.
- Adding angles to the wrong total (90° instead of 180°, or vice versa).
- Writing ray or angle names in the wrong order.

Geometry Basics Cheat Sheet

More Practice: measuring, naming, and angle relationships

1 SEGMENTS & MIDPOINTS

Segment Addition: AB + BC = AC,
when B is between A and C.
Example: AB = 8, BC = 11
so AC = 8 + 11 = 19.
A midpoint divides a segment into
two equal lengths.
If AC = 18 and B is the midpoint,
then AB = BC = 9.

2 DISTANCE & COORDINATES

A point in the plane is (x, y).
Horizontal distance: |x2 - x1|
Vertical distance: |y2 - y1|
Example: P(2, 3), Q(8, 3)
PQ = |8 - 2| = 6 units.
Distance formula:
d = sqrt((x2 - x1)^2 + (y2 - y1)^2)

3 ANGLE ADDITION

If ray BD lies inside angle ABC,
m(ABD) + m(DBC) = m(ABC).
Example: m(ABD) = 34°
and m(DBC) = 56°.
Then m(ABC) = 34° + 56°
= 90°.
The whole angle equals the sum
of its nonoverlapping parts.

4 COMPLEMENTARY & SUPPLEMEN

Complementary: two angles sum to 90°.
Example: 28° + 62° = 90°.
Supplementary: two angles sum to 180°.
Example: 115° + 65° = 180°.
A linear pair is always supplementary.
Vertical angles are always congruent.

5 POINT, LINE, PLANE

Point: exact location, no size.
Line: extends forever in two directions.
Plane: flat surface, extends forever.
Collinear: on the same line.
Coplanar: on the same plane.
Two intersecting lines meet at a point.
Perpendicular lines meet at 90°.

6 NAMING FIGURES

A segment has two endpoints.
Segment AB = segment BA.
A ray starts at its endpoint.
Ray AB starts at A; ray BA starts at B.
For angle ABC, B is the vertex.
Angle ABC = angle CBA.
Order matters for rays, not for segments.

7 QUICK REMINDERS & COMMON MISTAKES

- ✓ Name angles with the vertex in the middle.
- ✓ Use degree symbols when measuring angles.
- ✓ Midpoints split segments into equal halves.
- ✗ Do not confuse ray AB with ray BA.
- ✓ Sketch a figure and label points in order.
- ✗ Do not confuse congruent with supplementary.